Complete Java Methods Guide

String · Character · Array

Reference Tables · Code Examples · Interview Tips

■ 1. String Methods

Method	Description	Return	Example
charAt(int index)	Char at position	char	"hello".charAt(1) → 'e'
length()	String length	int	"hello".length() → 5
equals(Object)	Compare content	boolean	"hello".equals("hello") → true
equalsIgnoreCase(s)	Compare, ignore case	boolean	"Hello".equalsIgnoreCase("hello") → true
indexOf(String s)	First occurrence index	int	"hello".indexOf("l") → 2
lastIndexOf(String s)	Last occurrence index	int	"hello".lastIndexOf("l") → 3
substring(start)	Extract start→end	String	"hello".substring(2) → "llo"
substring(start,end)	Extract start to end-1	String	"hello".substring(1,4) → "ell"
split(String regex)	Split into array	String[]	"a,b,c".split(",") → ["a","b","c"]
toLowerCase()	To lowercase	String	"HELLO".toLowerCase() → "hello"
toUpperCase()	To uppercase	String	"hello".toUpperCase() → "HELLO"
trim()	Remove whitespace	String	" hello ".trim() → "hello"
contains(CharSequence)	Check if contains	boolean	"hello".contains("ell") → true
startsWith(String)	Check prefix	boolean	"hello".startsWith("he") → true
endsWith(String)	Check suffix	boolean	"hello".endsWith("lo") → true
isEmpty()	Check if empty	boolean	"".isEmpty() → true
replace(old, new)	Replace occurrences	String	"hello".replace("l","x") → "hexxo"
replaceAll(regex,new)	Replace via regex	String	"hello123".replaceAll("\\d","") → "hello"
concat(String str)	Append string	String	"hello".concat(" world") → "hello world"
toCharArray()	Convert to char[]	char[]	"hello".toCharArray() → [h,e,l,l,o]

■ String Methods — Code Example

```java
public class StringMethodsExample {
public static void main(String[] args) {
String str = " Hello World ";
String str2 = "Hello World";

// 1. Length & Trim
System.out.println(str.length()); // 15
System.out.println(str.trim()); // "Hello World"

// 2. Character access
System.out.println(str2.charAt(0)); // 'H'

// 3. Case conversion
System.out.println(str2.toUpperCase()); // "HELLO WORLD"
System.out.println(str2.toLowerCase()); // "hello world"

// 4. Comparison
System.out.println(str2.equals("Hello World")); // true
System.out.println(str2.equalsIgnoreCase("hello world")); // true
System.out.println(str2.contains("World")); // true

// 5. Index search
System.out.println(str2.indexOf("W")); // 6
System.out.println(str2.lastIndexOf("l")); // 9

// 6. Substring
System.out.println(str2.substring(0, 5)); // "Hello"
System.out.println(str2.substring(6)); // "World"

// 7. Split
String[] parts = "apple,banana,cherry".split(",");
System.out.println(parts[0]); // "apple"

// 8. Replace
System.out.println(str2.replace("World", "Java")); // "Hello Java"

// 9. Starts / Ends with
System.out.println(str2.startsWith("Hello")); // true
System.out.println(str2.endsWith("World")); // true

// 10. To char array
char[] chars = str2.toCharArray();
System.out.println(chars[0]); // 'H'
}
}
```

■ 2. Character Methods (Character Class)

Method	Description	Example
isLowerCase(char ch)	Check if lowercase	Character.isLowerCase('a') → true
isUpperCase(char ch)	Check if uppercase	Character.isUpperCase('A') → true
isDigit(char ch)	Check if digit	Character.isDigit('5') → true
isLetter(char ch)	Check if letter	Character.isLetter('a') → true
isWhitespace(char ch)	Check if whitespace	Character.isWhitespace(' ') → true
toLowerCase(char ch)	Convert to lowercase	Character.toLowerCase('A') → 'a'
toUpperCase(char ch)	Convert to uppercase	Character.toUpperCase('a') → 'A'
getNumericValue(char)	Numeric value of char	Character.getNumericValue('7') → 7
digit(char, radix)	Digit in given radix	Character.digit('F', 16) → 15
toString(char ch)	Convert to String	Character.toString('A') → "A"

■ Character Methods — Code Example

```
public class CharacterMethodsExample {
public static void main(String[] args) {
char ch1 = 'A';
char ch2 = 'a';
char ch3 = '5';
char ch4 = ' ';

// 1. Type checks
System.out.println(Character.isUpperCase(ch1)); // true
System.out.println(Character.isLowerCase(ch2)); // true
System.out.println(Character.isDigit(ch3)); // true
System.out.println(Character.isWhitespace(ch4)); // true
System.out.println(Character.isLetter(ch1)); // true

// 2. Case conversion
System.out.println(Character.toLowerCase(ch1)); // 'a'
System.out.println(Character.toUpperCase(ch2)); // 'A'

// 3. Numeric value
System.out.println(Character.getNumericValue('7')); // 7
System.out.println(Character.digit('F', 16)); // 15 (hex F = 15)

// 4. Convert to String
System.out.println(Character.toString(ch1)); // "A"

// 5. Unicode / ASCII value
System.out.println((int) ch1); // 65
}
}
```

■ 3. Array Methods (java.util.Arrays)

A. java.util.Arrays Class Methods

Method	Description	Example
`Arrays.sort(array)`	Sort array ascending	`Arrays.sort(arr)`
`Arrays.binarySearch(arr,key)`	Binary search (sorted arr required)	`Arrays.binarySearch(arr, 5)`
`Arrays.equals(arr1, arr2)`	Compare two arrays element-wise	`Arrays.equals(arr1, arr2)`
`Arrays.fill(array, value)`	Fill entire array with value	`Arrays.fill(arr, 0)`
`Arrays.copyOf(arr, newLen)`	Copy with new length (pad/truncate)	`Arrays.copyOf(arr, 10)`
`Arrays.copyOfRange(arr,f,t)`	Copy subrange [from, to)	`Arrays.copyOfRange(arr, 2, 5)`
`Arrays.toString(array)`	Readable string representation	`Arrays.toString(arr)` → `"[1,2,3]"`
`Arrays.asList(array)`	Convert to fixed-size List	`Arrays.asList(names)`
`Arrays.stream(array)`	Create a Stream from array	`Arrays.stream(arr).filter(x->x>3)`

B. Built-in Array Properties

Property / Syntax	Description	Example
`array.length`	Get array size (field, not method)	`arr.length` → 6
`for-each loop`	Iterate over every element	`for (int x : arr) { }`

■ Array Methods — Code Example

```java
import java.util.Arrays;

public class ArrayMethodsExample {
public static void main(String[] args) {
int[] arr = {5, 2, 8, 1, 9, 3};
int[] arr2 = {5, 2, 8, 1, 9, 3};
String[] names = {"Alice", "Bob", "Charlie"};

// 1. Sort
System.out.println(Arrays.toString(arr)); // [5, 2, 8, 1, 9, 3]
Arrays.sort(arr);
System.out.println(Arrays.toString(arr)); // [1, 2, 3, 5, 8, 9]

// 2. Binary search (array must be sorted first)
int idx = Arrays.binarySearch(arr, 5);
System.out.println("Index of 5: " + idx); // 3

// 3. Compare arrays
System.out.println(Arrays.equals(arr, arr2)); // false (arr2 not sorted)
Arrays.sort(arr2);
System.out.println(Arrays.equals(arr, arr2)); // true

// 4. Fill
int[] filled = new int[5];
Arrays.fill(filled, 7);
System.out.println(Arrays.toString(filled)); // [7, 7, 7, 7, 7]

// 5. Copy (extends to length 8 — pads with 0)
int[] copy = Arrays.copyOf(arr, 8);
System.out.println(Arrays.toString(copy)); // [1, 2, 3, 5, 8, 9, 0, 0]

// 6. Copy range → indices [1, 4)
int[] range = Arrays.copyOfRange(arr, 1, 4);
System.out.println(Arrays.toString(range)); // [2, 3, 5]

// 7. Array length
System.out.println(arr.length); // 6

// 8. For-each
for (int num : arr) {
System.out.print(num + " "); // 1 2 3 5 8 9
}

// 9. Convert to List
java.util.List<String> list = Arrays.asList(names);
System.out.println(list); // [Alice, Bob, Charlie]

// 10. As String
System.out.println(Arrays.toString(names)); // [Alice, Bob, Charlie]
}
}
```

■ Quick Interview Tips

- **String** is immutable — every modification returns a new object. Use `StringBuilder` for concatenation inside loops.
- **Character** methods are all *static* — always call via `Character.methodName()`.
- **Arrays.binarySearch()** requires a *sorted* array — call `Arrays.sort()` first.
- **Arrays.asList()** returns a *fixed-size* list — you cannot add or remove elements from it.